



Earthworms reproduce on Mars

Two worms were recently born from a colony living in soil created by NASA to simulate Mars dirt. <http://digit.in/MarsWorm>



Quantum simulator beats supercomputers

53 qubit quantum simulator can model physical interactions that are too complex for conventional supercomputers <http://digit.in/53qBit>

A tale of storms

The biggest, most unique storms and how they're getting bigger and more unpredictable

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If 2017 seemed like the year of storms to you, then you're not mistaken – this year in particular saw

quite a few high intensity storm events from around the world. Now, when it comes to storms and storm seasons in the different oceans of the world, there are quite a few parameters to measure and quantify them – ranging from peak wind speed to number of storms in a season to pressure at the eye of the storm and more. But, overall, these factors combine to make storms what they are today – deadly, frequent and somewhat unexpected. If you delve into the past of storm seasons around the world, the story isn't much different.

CHEATBOX

Just in case you're also confused about what's the difference between hurricanes, typhoons and cyclones, here's the thing - there's none. It's just about where they've originated:

- **Hurricane:** Atlantic and North-east Pacific
- **Typhoon:** Northwest Pacific
- **South Pacific and Indian Ocean:** Cyclones

TYPHOON TIP (1979)

When it comes to size and speed, Typhoon Tip ranks right up there on the top with some of the strongest storms ever. At its peak, its central pressure dropped to 870 mbar, which

is the lowest ever observed on Earth. This translated to wind gusts over 300kmph. And that's not even the most overwhelming fact about Tip – at its peak, it was 2,220km wide. To put some perspective on that – if it were to happen over India today, it would stretch from Delhi to Chennai and pretty much destroy everything in its path.

Fortunately for us mortals, the storm weakened before making landfall in Japan on Oct.19, 1979 – but it still was the most intense storm to hit the island of Honshu in almost a decade, killing more than 86 people and injuring hundreds. Even the city of Tokyo saw buildings swaying with the force of high-speed gusts.

TYPHOON NANCY (1961)

Nancy's rank among the top storms in the world isn't without controversy. Back in 1961, wind readings were most likely overestimated due to deficient technology and an inferior understanding of how typhoons work. But assuming that the readings were accurate, the numbers throw it right to the top with a peak wind-speed of 345kmph! It also qualifies for the record of the longest Category-5 storm in the Northern hemisphere yet – it

lasted in category 5 for five and a half days before making landfall.

When it made landfall in Japan, as a category 2, it caused more than \$500 million in damage and 200 deaths.

HURRICANE HARVEY (2017)

We've all heard of Hurricane Harvey this year. Part of the extremely active 2017 Atlantic Hurricane season, if Harvey's peak wind-speed of 215kmph and lowest pressure of 938 mbar don't

STORM STATS – WHAT DO THEY MEAN

Tropical storms are usually measured in a number of ways, but a couple of readings that you will mostly see are peak wind speeds and lowest central pressure. The first is measured at 10m above the earth's surface as an average of sustained wind speed over time (1min in US and 10min elsewhere) by satellite readings. Central pressure is measured by reconnaissance planes doing flybys over the storm and is also an indicator of whether the storm is growing or dying out.